

Understanding Asymmetric Collaboration in Augmented and Virtual Reality Immersive Environments

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Abstract

Asymmetric interaction designs improve collaboration and engagement in immersive learning. However, comparative studies between virtual reality (VR) and augmented reality (AR) are limited. This study examines how asymmetric interactions influence collaborative behaviors among unacquainted users in VR and AR. A controlled experiment with 52 participants was conducted, in which pairs completed puzzles incorporating informational, positional, and visual asymmetries. The results reveal that VR significantly improved immersion and flow, whereas AR promoted greater contextual awareness and natural coordination. Furthermore, VR was associated with increased cybersickness, while AR participants experienced greater disorientation. This study provides valuable information for optimizing collaborative learning and interaction design in immersive environments.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Virtual reality**; **Mixed / augmented reality**; • **Applied computing** → **Collaborative learning**.

Keywords

Collaborative Interaction, Virtual Reality, Augmented Reality, Interaction Design, Cybersickness

ACM Reference Format:

Lijie Zheng, Guoyueyang Cheng, Shaoteng Ke, Jiachen Yuan, Boon Giin Lee, and Matthew Pike. 2025. Understanding Asymmetric Collaboration

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758399>

in Augmented and Virtual Reality Immersive Environments. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758399>

1 Introduction

Virtual reality (VR) and augmented reality (AR) hold significant promise to enhance collaborative immersive learning [13]. Existing studies indicate that both VR and AR environments can individually improve engagement, teamwork, and learning outcomes [12]. Asymmetric interaction design, where collaborators have different information, roles, or perspectives, can further enrich these experiences [10]. Asymmetric configurations require active communication and diverse problem-solving approaches, ultimately increasing participation and team performance. The assignment of complementary asymmetric roles can alleviate initial discomfort and quickly foster trust among unfamiliar collaborators. However, systematic comparisons of how immersive VR qualities versus AR integration of real-world cues influence asymmetric collaboration remain limited, especially in contexts that require rapid social cohesion among strangers [16].

Asymmetric interaction designs, which assign varied roles or perspectives to team members, enhance engagement, trust, and collaboration among unfamiliar participants [3, 10]. Complementary role assignments mitigate initial social awkwardness and accelerate trust development [7]. However, existing comparative research primarily investigates symmetric VR-to-VR or VR-to-table collaborations, focusing on immersion rather than collaborative quality [1]. Even though VR environments maximize immersion and cognitive focus, they reduce real-world social cues. In contrast, AR supports natural interpersonal communication, but offers reduced immersive depth [9].

This paper introduces *CritterEscape*, an immersive collaborative puzzle game with asymmetric roles in both VR and AR, based on the foundational work of [8]. This study expanded the design that enables a direct comparison of how VR and AR modalities affect user experience and cooperative behavior, allowing for systematic observation of how each medium's characteristics modulate social

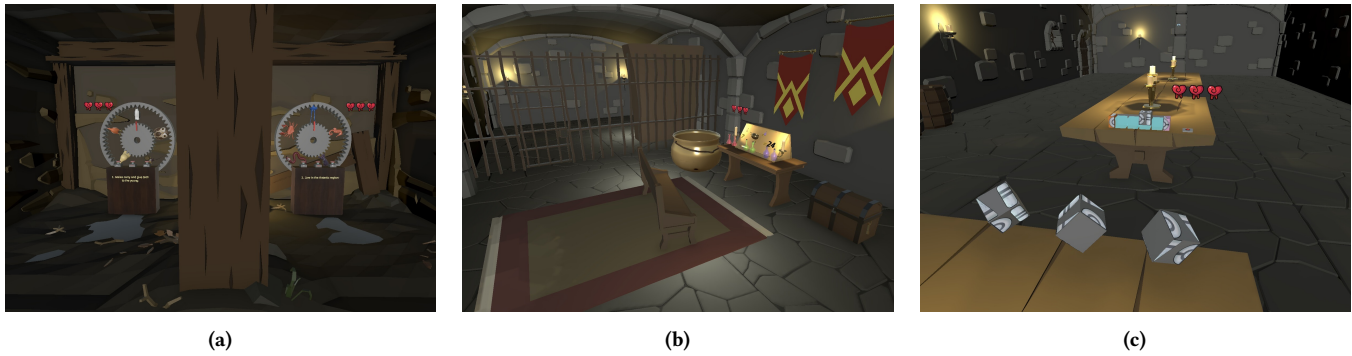


Figure 1: Overview puzzle designs integrate three key features: (a) informational asymmetry, which requires mutual information exchange; (b) spatial asymmetry, which emphasizes positional cooperation; and (c) visual asymmetry, which necessitates precise visual communication.

interaction under consistent task conditions. This study focuses on two core research questions:

- (1) How do asymmetric designs in collaborative puzzles influence social interaction among unacquainted users in VR and AR?
- (2) How do the experiences of unacquainted users differ in collaborative asymmetric games between VR and AR?

By comparing VR and AR in identical scenarios, this study helps identify which immersive platform and asymmetry strategies most effectively facilitate initial ice-breaking and team cohesion for unfamiliar collaborators.

2 Methodology and Experiment Design

The game design incorporated three asymmetric interaction mechanisms: informational, positional, and visual asymmetries, as depicted in Figure 1. Information asymmetry, based on social interdependence theory [6], required collaborators to verbally exchange complementary clues to collaboratively solve an animal identification puzzle. Positional asymmetry, based on constructivist learning theory [4], spatially separated collaborators within a potion-brewing task to stimulate active spatial communication and cooperation. Finally, visual asymmetry used principles from cognitive load theory [15]; collaborators, presented with different visual perspectives, required precise verbal articulation to solve a geometric alignment puzzle. Each puzzle maintained identical task logic, timing, scoring, and risk elements in VR and AR modalities, isolating the effects of immersive technology on collaborative behaviors and experiences.

The between-subject experiments were conducted with 52 participants (aged 18 to 26, $M = 21.0$, $SD = 1.8$). The participants were randomly paired and assigned VR or AR conditions, with the condition that both participants were unfamiliar to each other. All participants had no experience with immersive VR or AR games. After a brief tutorial on equipment usage, pairs collaboratively completed the mini-games. The post-experiment evaluations included validated questionnaires measuring the user experience of the gameplay (GEQ; [5]), social interdependence (SOCS; [14]), collaborative learning and social presence (CLSS; [11]) and cybersickness (SSQ; [2]). In addition, semi-structured interviews provided qualitative

information on participant experiences, collaborative dynamics, and perceived discomfort. This comprehensive mixed-methods approach ensured robust data, capturing both quantitative metrics and qualitative user feedback.

3 Results and Discussion

The results show that VR improved immersion, flow, and engagement, but increased perceived challenge and cybersickness. AR facilitated natural social coordination, operational competence, and rapid mutual understanding, but increased emotional sensitivity and potential distraction. Both VR and AR supported social cohesion and fundamental collaborative experiences, emphasizing the critical importance of task design. Practical implications suggest VR for tasks demanding intensive cognitive engagement and AR for scenarios prioritizing intuitive and socially grounded interactions.

Quantitative analysis revealed significant differences between VR and AR conditions. Participants in VR condition reported a particularly higher immersion ($p < 0.001$), flow experience ($p < 0.001$), and perceived challenge ($p = 0.001$). In contrast, AR participants reported higher levels of competence ($p = 0.021$), suggesting more intuitive task interactions. Post-game evaluations indicated that VR participants had greater difficulty returning to reality ($p < 0.001$). Furthermore, participants in VR condition experienced significantly higher cybersickness symptoms, particularly nausea ($p < 0.001$) and oculomotor discomfort ($p < 0.002$).

These findings highlight the distinct strengths and limitations of VR and AR for collaborative asymmetric interactions. VR's immersive qualities effectively support tasks requiring deep cognitive engagement and high focus, but its potential for cybersickness requires careful management. AR's capacity for contextual awareness and natural social interaction makes it particularly valuable for initial collaboration scenarios, despite potential distractions and increased emotional sensitivity. SOCS and CLSS results indicated that both modalities similarly support fundamental collaborative experiences, highlighting the importance of thoughtful task design.

4 Conclusion

This study empirically examined how informational, positional, and visual asymmetry shape collaborative experiences differently

in VR and AR environments. VR demonstrated superior immersion and focused cognitive engagement, while AR facilitated natural interpersonal interactions and faster social coordination. These findings offer actionable guidance for educators and designers in effectively utilizing each platform's unique strengths to enhance collaborative learning. However, several limitations were observed in this study. This study explored only three asymmetric mechanics; future studies could investigate broader forms of asymmetry, such as temporal variations or multi-role interactions. Relying on self-reported measures poses potential biases; integrating objective physiological, gaze, and cognitive load measures would improve validity. Finally, no detailed analysis of the dynamics of group interaction was performed, including leadership roles and discourse patterns. Future studies including multi-modal analysis could further clarify in-depth underlying collaborative mechanisms.

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